PLSC 476: Empirical Legal Studies

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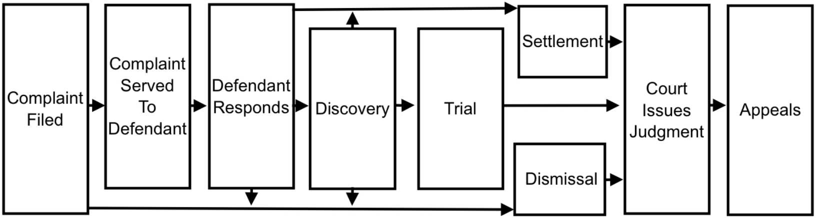
The Civil Litigation Process



Civil Litigation: Empirical Moments

- Document Review (pre-litigation; for contracts, etc.)
- Legal Research (finding relevant laws, precedents, etc.)
- Predictive Tools (litigation costs, likely trial outcomes, settlement probabilities, etc.)
- Settlement Calculators
- Electronic Discovery
- Many others...

Relevant Federal Law

- The *Stored Communication Act* ("SCA") (1986)
  - Foundational law governing access to electronically stored information held by third parties (especially ISPs)
  - Both protects users from unlawful access to their data *and* defines the terms under which compelled disclosure of that data may take place
  - Also governs data preservation
- The *CLOUD* Act ("Clarifying Lawful Overseas Use of Data") (2018)
  - Extends the SCA to data held outside the U.S.
- The *Federal Rules of Civil Procedure* ("FRCP")
  - The "First Rule" of civil procedure: The FRCP are "construed, administered, and employed by the court and the parties to secure the just, speedy, and inexpensive determination of every action and proceeding."
  - Amended 2006, 2009 & 2015 to cope with electronically stored information ("ESI") and e-discovery

How Discovery Works: A Brief History

???-1960s: Paper-Based Discovery

1. Attorneys request (paper) copies of relevant documents
2. Attorneys review documents manually for relevance

1970s-2010s: eDiscovery 1.0

1. Attorneys request electronic (image or OCR-ready) copies of relevant documents
2. Attorneys review documents for relevance
   a. 1970s-1990s: Manually
   b. 1990s-2010s: Digitally (via search tools)

2010s-Present: eDiscovery 2.0 ("Technology-Assisted Review")

1. Attorneys request machine-readable electronic documents
2. Legal services companies (LSCs) use machine learning / text analysis / LLM tools to review documents
3. Attorneys review LSC findings

Text As Data: Concepts

Machine Learning ("ML")

- Teaching computers to "think" and learn like humans
- Includes a wide range of classification and prediction models (including simple multivariate ones like we used last week)
- Uses: Search + recommender engines; image/voice recognition; forecasting; others

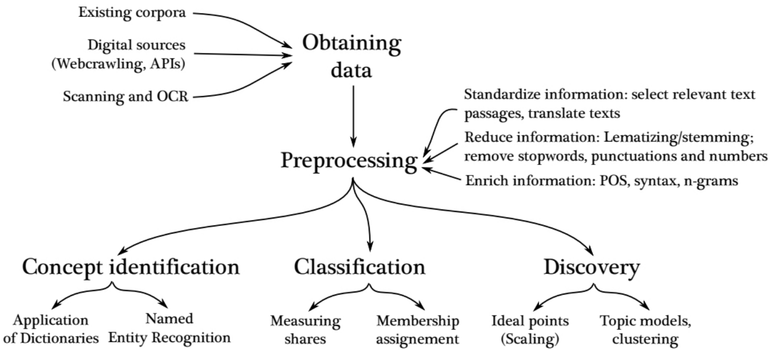
Natural Language Processing ("NLP")

- How computers "understand" human language
- Generally requires text to be represented numerically
- Uses: Predictive text (spell checking, mail filters, etc.); machine translation; attribution/plagiarism detection; others

Flavors:

- Supervised:
    - Train a computer to recognize / predict patterns in data **A**
    - $\rightarrow$ use that training to identify patterns in data **B**
- Unsupervised: Use patterns in the data to group/classify text/documents

Text As Data: Processes



Source

Stupid Text Tricks: Semantic Analysis

Variations ("S" = supervised, "U" = unsupervised):

- Part-Of-Speech Tagging (S + U)
- Word Sense Disambiguation (U)
- Named Entity Recognition (S + U)
- Sentiment Analysis (S + U)
- Topic Modeling (S + U)
- Terminology Extraction (U)
- Large Language Models (U)

How Do We Analyze Text?

General Idea:

- Represent text $\mathfrak{D}$ as a numerical array $\mathbf{C}$
- Analyze $\mathbf{C}$ $\rightarrow$ generate predictions / classifications $\hat{\mathbf{P}}$
- Map $\hat{\mathbf{P}}$ back to $\mathfrak{D}$ or $\mathfrak{D}'$

Key Steps / Concepts:

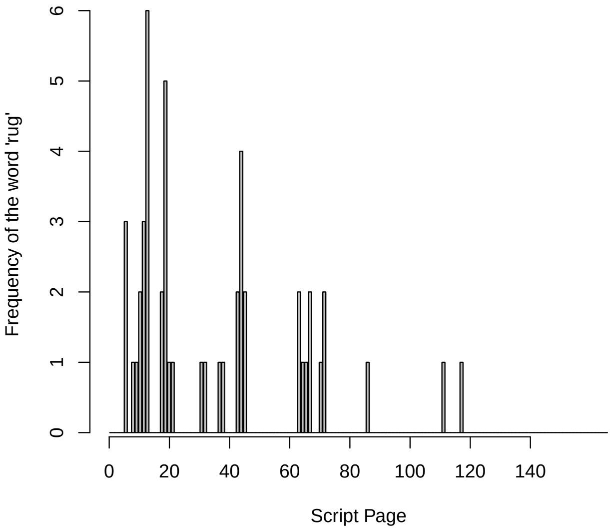
1. Text *Preprocessing*:
    - Removing capitalization, punctuation, "stop words," etc.
    - Stemming / lemmatization (e.g., `traveler`, `traveling` $\rightarrow$ `travel*`)
2. *Document-Term Matrix* ("DTM")
    - Rows = *documents* (sentences, speeches, tweets, etc.)
    - Columns = *terms* (words / phrases)
    - Cells = counts (or weighted counts)

An Example...

Creating a DTM

```
> TBL<-pdf_text("https://inwardeleven.com/lebowski/The%20Big%20Lebowski.pdf")

> # Turn that into a "corpus" + term-document matrix:

> TBL.corp<-SimpleCorpus(VectorSource(TBL))
>
> TBL.dtm <- DocumentTermMatrix(TBL.corp,
+                control=list(removePunctuation=TRUE,
+                             stopwords=TRUE,
+                             tolower=TRUE,
+                             stemming=TRUE,
+                             removeNumbers=FALSE,
+                             weight=weightTfIdf))

> dim(TBL.dtm)
[1]  138 3011

> inspect(TBL.dtm)

<<DocumentTermMatrix (documents: 138, terms: 3011)>>
Non-/sparse entries: 9549/405969
Sparsity           : 98%
Maximal term length: 16
Weighting          : term frequency (tf)
Sample             :
     Terms
Docs  cont'd continu dude fuck know lebowski look man maud walter
  100      0       3    6    0    0        0    2   0    3      0
  101      0       2    6    0    2        1    1   1    0      0
  114      0       2    2    2    1        0    2   7    0      1
  2        1       1    4    0    0        2    1   2    0      0
  36       0       2    8    3    1        0    3   3    0      3
  37       0       2    8    0    0        0    3   2    0      0
  4        2       3   10    0    0        3    0   0    0      0
  69       1       2    8    0    0        5    1   8    0      0
  7        0       2    5    0    0        0    1   5    0      3
  93       1       2    6    2    1        1    2   2    0      3
```

Frequency of the Word “rug”



Terms Correlated with the Word "rug"